

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OPERATING SYSTEMS COURSE CODE: CSA04

COURSE OUTCOME COVERED

C02: Analyze process management and deadlock handling techniques to improve CPU utilization and system performance(BL3-BL4)

Assessment Tool Weightage: 30%

Dr. M. Anoop

QUESTIONS: 10

Total Marks:50

Annexure A

Scenario based

Code of Conduct:

I Srinivas S certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1 — CPU Scheduling in a Real-Time Ticket Booking System

A railway ticket booking system experiences heavy load during festival seasons. The system handles three categories of processes:

- P1 – Payment gateway transactions (High priority)
- P2 – Seat availability checks (Medium priority)
- P3 – Ticket confirmation and printing (Low priority)
- Burst times vary from 2 ms to 20 ms.

Recently, customers complained that ticket confirmations are taking too long.

Tasks

1. Identify the most suitable scheduling algorithm (FCFS, SJF, Priority, RR) for this scenario.
2. Justify *why* your chosen method improves CPU utilization and reduces customer wait time.
3. Provide a sample execution order for the above categories.
4. Discuss one drawback of this scheduling method in real-time systems.

(10 Marks)

A.

1. Most suitable scheduling algorithm

Priority Scheduling

2. Justification

- Payment transactions (P1) are most critical and should execute first.
- Seat availability (P2) is next in importance.
- Ticket confirmation (P3) can execute after higher-priority tasks.
- Priority scheduling reduces waiting time for critical tasks and improves CPU utilization by serving important processes immediately.

3. Sample execution order

P1 (Payment)

↓

P2 (Seat Availability)

↓

P3 (Ticket Confirmation & Printing)

4. Drawback

Low-priority processes may suffer from starvation if high-priority processes continue to arrive.

Question 2 — Deadlock in a Banking Transaction Server

A bank server handles the following operations simultaneously:

- T1: Transfers money between accounts
- T2: Updates passbook entries
- T3: Generates monthly statements

Due to simultaneous locking of Account Database (A), Customer Info Database (B), and Transaction Logs (C), the system enters a state where:

- T1 holds A and requests B
- T2 holds B and requests C
- T3 holds C and requests A

Tasks

1. Draw the resource request situation (text-based RAG).
2. Identify whether a deadlock has occurred and justify your answer.
3. Recommend *one specific technique* (avoidance/prevention/recovery) and explain how it fixes the issue.
4. Suggest a redesign for the locking order to avoid future deadlocks.

(10 Marks)

A.

1. Text-based Resource Allocation Graph (RAG)

T1 → B

↑

A

T2 → C

↑

B

T3 → A

↑

C

Or simply,

T1 holds A → requests B

T2 holds B → requests C

T3 holds C → requests A

2. Has deadlock occurred?

Yes.

Reason:

A circular wait exists.

T1 → B

T2 → C

T3 → A

Each process waits for a resource held by another process.

3. Recommended technique

Deadlock Prevention

Use a fixed order for requesting resources.

Example:

Always acquire:

A → B → C

This removes circular waiting.

4. Locking redesign

Every transaction should request resources only in this order:

Account Database (A)

↓

Customer Database (B)

↓

Transaction Log (C)

No process should violate this sequence.

Question 3 — Mobile OS App Switching Lag

A mobile operating system is designed to support smooth app switching.

But users report that:

- Opening WhatsApp delays the background music player.
- Switching between social media apps feels laggy.
- CPU shows high idle times despite multiple running apps.

The OS currently uses Round Robin (quantum = 20 ms) with 20 background apps active.

Tasks

1. Analyze why the CPU remains idle despite many active processes.
2. Determine whether the time quantum is helping or hurting performance.
3. Recommend a better scheduling method or quantum value, with justification.
4. Explain how the change will reduce switching lag and improve system throughput.

(10 Marks)

A.

1. Why CPU remains idle?

- Many apps are waiting for I/O operations.
- Some apps are sleeping in the background.
- The scheduler has no ready processes during I/O wait periods.

2. Is 20 ms quantum helping?

No.

A 20 ms quantum is relatively large for mobile applications.

Effects:

- Slow response
- Delayed app switching
- Poor interactive performance

3. Better scheduling recommendation

Use

- **Round Robin with 10 ms quantum, or**
- **Multilevel Feedback Queue (MLFQ)**

Reason:

- Better responsiveness
- Faster switching between foreground apps
- Higher throughput

4. Improvement

- Faster context switching
- Better CPU utilization
- Reduced app-switch delay
- Improved user experience

Question 4 — Deadlock Risk in an Operating System Update

During a Windows OS update, several modules run in parallel:

- M1: Copy update files
- M2: Install drivers
- M3: Update registry
- M4: Validate system integrity

Resources involved:

- R1 = Disk I/O
- R2 = Kernel lock
- R3 = Configuration lock

Recently logs show:

- M1 holds R1, waiting for R2
- M2 holds R2, waiting for R3
- M3 holds R3, waiting for R1

Tasks

1. Identify whether this sequence can lead to a deadlock.
2. If yes, explain the exact circular wait condition.
3. Propose *Banker's algorithm style* resource allocation to avoid risk.
4. Suggest how OS designers can prioritize modules to improve CPU utilization during update.

(10 Marks)

A.

1. Can this lead to deadlock?

Yes.

2. Circular wait condition

M1 holds R1 → waits for R2

M2 holds R2 → waits for R3

M3 holds R3 → waits for R1

Cycle:

M1 → R2

M2 → R3

M3 → R1

Hence deadlock occurs.

3. Banker's Algorithm approach

Grant resources only if the system remains in a **safe state**.

Example:

- Allocate R2 to M1 only if enough resources remain for other modules to complete.
- Otherwise, delay allocation.

This prevents unsafe states.

4. Prioritization strategy

Priority order:

M1 (Copy files)

↓

M2 (Install drivers)

↓

M3 (Update registry)

↓

M4 (Validation)

Benefits:

- Better CPU utilization
- Less waiting
- Faster completion
- Reduced deadlock risk

Question 5 — Cloud Server CPU Utilization Analysis

A cloud hosting provider notices:

- 80% of the time, VM processes wait for network or disk
- Only 20% of the time is spent on CPU
- Each server runs 7 VMs

Using the formula:

$$U = 1 - p^n$$

where p = I/O wait probability

Tasks

1. Calculate CPU utilization.
2. Interpret whether the server is underutilized or optimally loaded.
3. Suggest how increasing or decreasing number of VMs affects throughput.
4. Propose two OS-level strategies for improving CPU utilization in this cloud environment.

(10 Marks)

A.

RUBRICS FOR CALCULATION

Scenario based assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand